Statistical Arbitrage via Pairs Trading: Algorithm Development, Mathematical Modelling, and Backtesting

Bachelor Thesis Project – II report submitted to

Indian Institute of Technology Kharagpur

in partial fulfilment for the award of the degree of

Bachelor of Technology

in

Mechanical Engineering

by

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(22ME3FP49)

Under the supervision of

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Spring Semester, 2025-26

April 29, 2026

DECLARATION

I certify that

(a) The work contained in this report has been done by me under the guidance of my supervisor.

(b) The work has not been submitted to any other Institute for any degree or diploma.

(c) I have conformed to the norms and guidelines given in the Ethical Code of Conduct of the Institute.

(d) Whenever I have used materials (data, theoretical analysis, figures, and text) from other sources, I have given due credit to them by citing them in the text of the thesis and giving their details in the references. Further, I have taken permission from the copyright owners of the sources, whenever necessary.

Date: April 29, 2026	(Uday om srivastava)
Place: Kharagpur	(22ME3FP49)

DEPARTMENT OF MECHANICAL ENGINEERING

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CERTIFICATE

This is to certify that the project report entitled "**Statistical Arbitrage via Pairs Trading: Algorithm Development, Mathematical Modelling, and Backtesting**" submitted by **Uday om srivastava** (Roll No. 22ME3FP49) to Indian Institute of Technology Kharagpur towards partial fulfilment of requirements for the award of degree of Bachelor of Technology in Mechanical Engineering is a record of bona fide work carried out by him under my supervision and guidance during Spring Semester, 2025-26.

Date: April 29, 2026

Place: Kharagpur

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Abstract

Name of the student: **Uday om srivastava** Roll No: **22ME3FP49**

Degree for which submitted: **Bachelor of Technology**

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Thesis supervisor: **Prof. Arindam Banerjee**

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This thesis presents a complete pairs trading system applied to S&P 500 equities. Pairs trading is a market-neutral statistical arbitrage strategy that exploits temporary divergence and subsequent mean reversion of two co-integrated assets.

The work rests on three pillars. First, an unsupervised machine-learning pipeline – comprising PCA dimensionality reduction and OPTICS density-based clustering – narrows the S&P 500 universe to statistically robust candidate pairs, validated through cointegration testing, the Hurst exponent, OU half-life, and mean-cross frequency filters. Second, four mathematically distinct trading strategies are derived from first principles: Bollinger Bands, the Ornstein-Uhlenbeck (OU) stochastic-process model with maximum-likelihood calibration, a Copula-based conditional-probability approach (Clayton, Gumbel, Frank), and a Cointegration strategy with dynamic hedge-ratio estimation. Third, all strategies are backtested out-of-sample during a significant S&P 500 drawdown (2022–2023). Across three pairs – AEP/CNP, AEE/AEP, and

LNT/CNP – all strategies generated positive returns (4%–19.5%) while SPY returned −2.87% with a −22.09% maximum drawdown. The Copula model consistently delivered the highest absolute returns; Bollinger and Cointegration achieved the best risk-adjusted returns (Sharpe up to 2.087). The full implementation is open-source at `https://github.com/ImaginedTime/Pairs_trading_BTP2`.

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Chapter 1

Introduction

1.1 Background and Motivation

Financial markets attract systematic strategies that generate returns independent of broad market movements. Directional approaches such as momentum trading can deliver strong returns in trending markets but suffer severe drawdowns during reversals. *Pairs trading* offers an alternative: it is a market-neutral statistical arbitrage strategy that exploits the temporary divergence between two historically co-moving assets. A simultaneous long position in the underpriced asset and a short position in the overpriced asset profits once the spread reverts to its mean, while net exposure to broad market moves is largely cancelled out.

Pairs trading was first applied systematically by Nunzio Tartaglia's group at Morgan Stanley in the 1980s [5]. Since then, a rich literature has developed around the mathematical underpinnings of pair selection, spread modelling, and trade execution. This project implements a complete pipeline on S&P 500 equities, drawing inspiration from Bhagat et al. [2] while contributing an independent, modular Python implementation with an extended model comparison.

1.2 Objectives

1. Design an *unsupervised machine-learning pipeline* for pair selection using PCA and OPTICS clustering.

2. Derive and implement *four trading strategies* – Bollinger Bands, Ornstein-Uhlenbeck (OU), Copula, and Cointegration – each grounded in rigorous statistical theory.
3. Build a *realistic backtesting engine* accounting for hedge-ratio-based position sizing and transaction costs.
4. Empirically *compare all strategies* across multiple pairs against a SPY buy-and-hold benchmark.
5. Release the work as a *reproducible Python package*.

1.3 Scope

The project uses daily closing prices of S&P 500 equities from Stooq. Strategy parameters are estimated on a training window (up to February 2022) and evaluated on a strictly out-of-sample test period (March 2022 – March 2023). Intraday execution, options, and portfolio-level multi-pair allocation are outside scope.

1.4 Report Organisation

Chapter 2 surveys the relevant literature. Chapter 3 details the data pipeline and pair-selection methodology. Chapter 4 derives the mathematics of each trading strategy. Chapter 5 presents backtesting results. Chapter 6 concludes with a discussion and remarks.

Chapter 2

Literature Survey

2.1 Pairs Trading

Pairs trading traces its academic roots to Gatev, Goetzmann, and Rouwenhorst [5], who documented annualised excess returns of approximately 11% on U.S. equities (1962–2002) using a distance-based pair-selection method. Vidyamurthy [10] provided the theoretical framework, placing pairs trading within *statistical arbitrage* and formalising the mean-reversion requirement via cointegration theory.

Later work shifted pair selection from distance metrics to formal statistical tests. Engle and Granger [4] showed that two individually non-stationary (I(1)) series can share a stationary linear combination – a property called *cointegration* – that guarantees the spread will not drift permanently. This is a far stronger condition than simple correlation. More recent approaches use unsupervised learning to cluster assets before testing cointegration, reducing the $O(n^2)$ search space significantly; OPTICS [1] is particularly well-suited because it does not require the number of clusters to be pre-specified and handles density-heterogeneous financial data well.

2.2 Spread Modelling

Ornstein-Uhlenbeck process. The OU process is the canonical continuous-time model for a mean-reverting signal. Elliott et al. [3] showed how its parameters can be estimated via maximum likelihood and used to construct trading thresholds with provable optimality properties. Leung and Li [6] extended this to optimal entry/exit timing under the OU model.

Copula-based models. Liew and Wu [7] argued that spread-based frameworks assume linear dependence, whereas copulas (Sklar [8]) capture the full nonlinear dependence structure. Trading signals derived from conditional copula probabilities are theoretically superior because they are invariant to marginal distribution misspecification. Stander et al. [9] found that Archimedean copulas (Clayton, Gumbel, Frank) better capture equity tail dependence than the Gaussian copula.

2.3 Mean Reversion Testing

The Hurst exponent H measures the long-memory character of a time series: $H < 0.5$ indicates anti-persistence (mean reversion), $H = 0.5$ a random walk, and $H > 0.5$ trending behaviour [11]. Requiring $H < 0.5$ on the spread is a necessary condition for profitable pairs trading [10].

2.4 Backtesting

In this project, all parameters are estimated exclusively on training data and strategies are evaluated on a strictly out-of-sample test period with no look-ahead bias.

Chapter 3

Data and Pair Selection Pipeline

3.1 Data and Preprocessing

Daily OHLCV bars for S&P 500 constituents are downloaded from the Stooq data feed and cached locally (one CSV per ticker under `data/raw/`) to ensure reproducibility. Tickers covering fewer than 80% of calendar trading days in the training window are excluded. The cleaned data is aligned into a price matrix $\mathbf{P} \in \mathbb{R}^{T \times N}$ where T is the number of trading days and N the number of surviving tickers.

Daily log-returns are then computed as

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right), \tag{3.1}$$

and standardised column-wise (zero mean, unit variance) to put all assets on equal footing.

3.2 Dimensionality Reduction via PCA

The return matrix $\mathbf{R} \in \mathbb{R}^{(T-1) \times N}$ is high-dimensional and noisy. PCA projects it onto the k^* directions of maximum variance, where k^* is the smallest integer such that the cumulative explained variance ratio reaches 80%:

$$\mathrm{EVR}(k^*) = \frac{\sum_{j=1}^{k^*} \lambda_j}{\sum_{j=1}^{N} \lambda_j} \geq 0.80. \tag{3.2}$$

The projected feature matrix $\widetilde{\mathbf{X}} = \mathbf{R}\,\mathbf{V}_{k^*} \in \mathbb{R}^{(T-1)\times k^*}$ retains the dominant market factors while discarding noise, making the subsequent clustering more reliable. PCA is fit only on training data; the same projection is applied to test data without refitting.

3.3 Clustering via OPTICS

OPTICS [1] orders data points by their *reachability distance*:

$$\text{reach-dist}(p, q) = \max\bigl(\text{core-dist}(q),\ d(p, q)\bigr), \tag{3.3}$$

where $d(p, q)$ is the Euclidean distance in the PCA-reduced space and core-dist(q) is the distance from q to its `min_samples`-th nearest neighbour. Valleys in the resulting reachability plot correspond to dense clusters. With `min_samples`$= 3$, OPTICS identified **5 clusters** from the S&P 500 universe, yielding **34 candidate pairs** – a 97% reduction from the unconstrained $\binom{500}{2}$ pair space.

3.4 Statistical Filtering

Each candidate pair must pass all four filters below to advance to the strategy stage.

Cointegration (Engle-Granger). The hedge ratio $\hat{\beta}$ is estimated by OLS regression of P^A on P^B, forming the spread residual $Z_t = P_t^A - \hat{\beta} P_t^B$. An ADF test on Z_t must yield $p < 0.05$, confirming stationarity.

Hurst Exponent. Via rescaled-range analysis, $\hat{H}$ is estimated from $\mathbb{E}[R(n)/S(n)] \propto n^H$. The criterion $\hat{H} < 0.5$ confirms anti-persistence (mean reversion) of the spread.

Half-Life of Mean Reversion. Regressing $\Delta Z_t = \phi Z_{t-1} + \eta_t$ gives $t_{1/2} = -(\ln 2)/\ln(1+\hat{\phi})$. The criterion $1 \leq t_{1/2} \leq 252$ trading days ensures the reversion is neither too noisy nor too slow to trade.

Mean Crosses Per Year. A minimum of 12 zero-crossings per year of the demeaned spread ensures sufficient trading opportunities.

3.5 Selected Pairs by Backtest Year

The pipeline is run independently for each test year, with the training window being the full 12 months immediately preceding the test period. Table 3.1 summarises the pairs selected for each year alongside their key selection statistics from the CSV outputs.

Table 3.1: Selected pairs for each backtesting period with statistical filter metrics.

Test Year	Pair	Asset A	Asset B	Coint. p	$\hat{\beta}$	Hurst H	Half-life (d)	Mean X/yr
2022	AEP/CNP	American Electric Power	CenterPoint Energy	0.0023	0.294	0.322	6.8	35.9
	AEE/AEP	Ameren Corporation	American Electric Power	0.0082	1.029	0.265	6.2	33.1
	AWK/ATO	American Water Works	Atmos Energy	0.0229	0.457	0.403	8.0	21.2
	AEE/CNP	Ameren Corporation	CenterPoint Energy	0.0262	0.307	0.410	10.0	19.3
	LNT/CNP	Alliant Energy	CenterPoint Energy	0.0422	0.321	0.475	13.2	34.0
2023	AWK/CNP	American Water Works	CenterPoint Energy	0.0124	0.080	0.393	7.2	21.4
	AEP/CNP	American Electric Power	CenterPoint Energy	0.0316	0.156	0.379	9.4	27.9
	LNT/AWK	Alliant Energy	American Water Works	0.0367	4.357	0.408	8.0	28.8
	LNT/CNP	Alliant Energy	CenterPoint Energy	0.0437	0.448	0.316	7.3	34.4
2024	APO/BAC	Apollo Global Management	Bank of America	0.0124	0.135	0.404	8.1	29.8
	ARES/BAC	Ares Management	Bank of America	0.0364	0.163	0.429	9.0	27.0

A notable observation is the shift in sector composition across test years. The 2022 pairs are entirely from the U.S. utilities sector (AEP, CNP, AEE, LNT), consistent with their shared regulatory and interest-rate exposure. By 2024, the pipeline selected pairs from the financial services sector (APO, BAC, ARES), reflecting a shift in which sectors exhibited the strongest statistical co-movement during that training window. This illustrates the data-driven nature of the ML selection pipeline: it adapts automatically to market structure without requiring any manual sector classification.

Chapter 4

Trading Strategies

For a pair (A, B) with OLS hedge ratio $\hat{\beta}$, the *spread* is

$$Z_t = P_t^A - \hat{\beta}\, P_t^B. \tag{4.1}$$

A signal of $+1$ means *long the spread* (long A, short $\hat{\beta}$ units of B); -1 means *short the spread*; 0 means flat.

4.1 Bollinger Band Strategy

Let $w = 20$ days and $k = 2$. Define the rolling mean and standard deviation of Z_t:

$$\mu_t^{(w)} = \frac{1}{w} \sum_{s=t-w+1}^{t} Z_s, \qquad \sigma_t^{(w)} = \sqrt{\frac{1}{w-1} \sum_{s=t-w+1}^{t} (Z_s - \mu_t^{(w)})^2}, \tag{4.2}$$

and the rolling z-score $z_t = (Z_t - \mu_t^{(w)})/\sigma_t^{(w)}$. The signal rule is:

$$\text{signal}_t = \begin{cases} -1 & z_t > k \quad \text{(spread above upper band: short)}, \\ +1 & z_t < -k \quad \text{(spread below lower band: long)}, \\ 0 & |z_t| < \varepsilon_{\text{exit}} \quad \text{(mean cross: exit)}, \\ \text{hold} & \text{otherwise.} \end{cases} \tag{4.3}$$

4.2 Ornstein-Uhlenbeck Strategy

4.2.1 The OU Model

The spread is modelled as an Ornstein-Uhlenbeck process:

$$dZ_t = \kappa(\theta - Z_t)\,dt + \sigma\,dW_t, \tag{4.4}$$

where θ is the long-run mean, $\kappa > 0$ the speed of mean reversion, $\sigma > 0$ the volatility, and W_t a standard Brownian motion. The stationary distribution is $Z_t \sim \mathcal{N}(\theta,\, \sigma^2/2\kappa)$, and the half-life of reversion is $t_{1/2} = \ln 2/\kappa$.

4.2.2 Exact Discrete Transition and MLE

Solving (4.4) analytically for $\Delta t = 1$ day yields

$$Z_t \mid Z_{t-1} \sim \mathcal{N}\Big(\theta + (Z_{t-1} - \theta)e^{-\kappa},\ \ \frac{\sigma^2}{2\kappa}(1 - e^{-2\kappa})\Big). \tag{4.5}$$

The log-likelihood for observations $\{Z_0, \ldots, Z_T\}$ is

$$\ell(\kappa, \theta, \sigma) = -\frac{T}{2}\ln(2\pi\sigma_\varepsilon^2) - \frac{1}{2\sigma_\varepsilon^2}\sum_{t=1}^{T}\big[Z_t - \theta(1-\alpha) - \alpha Z_{t-1}\big]^2, \tag{4.6}$$

where $\alpha = e^{-\kappa}$ and $\sigma_\varepsilon^2 = \sigma^2(1-\alpha^2)/(2\kappa)$. Parameters (κ, θ, σ) are estimated by numerically minimising $-\ell$ via L-BFGS-B. Additionally, the optimal weight $w^* \in [0.01, 1]$ for the portfolio $Z_t(w) = wP_t^A + (1-w)P_t^B$ is found by maximising ℓ over a grid search.

4.2.3 Signal Generation

With the equilibrium z-score $s_t = (Z_t - \theta)/\sqrt{\sigma^2/2\kappa}$, entry and exit thresholds mirror the Bollinger approach but are now grounded in the *stationary* distribution of the OU process:

$$\text{signal}_t = \begin{cases} -1 & s_t > 2, \\ +1 & s_t < -2, \\ 0 & |s_t| < 0.5, \\ \text{hold} & \text{otherwise.} \end{cases} \tag{4.7}$$

For multi-step forecasting, the Euler-Maruyama scheme $Z_{t+1} = Z_t + \kappa(\theta - Z_t) + \sigma\xi_t,\ \xi_t \sim \mathcal{N}(0, 1)$ is simulated with 10,000 paths; the 98% confidence interval serves as adaptive bands.

4.3 Copula Strategy

4.3.1 Theoretical Foundation

Rather than modelling the spread directly, the Copula approach captures the full dependence structure between asset returns. Let $u_t = F_A(r_t^A)$ and $v_t = F_B(r_t^B)$ be the probability integral transforms (ranks) of the daily returns with respect to their empirical CDFs. By Sklar's theorem [8], the joint distribution factors as $F_{AB}(r^A, r^B) = C(F_A(r^A), F_B(r^B))$, where $C : [0, 1]^2 \to [0, 1]$ is the copula encoding the dependence separately from the marginals. Trading signals come from the conditional probabilities

$$p^A = C(u \mid v) = \frac{\partial C(u, v)}{\partial v}, \qquad p^B = C(v \mid u) = \frac{\partial C(u, v)}{\partial u}. \tag{4.8}$$

4.3.2 Copula Families

The copula parameter θ is estimated from Kendall's τ via family-specific closed-form inversions. Three Archimedean families are implemented:

Clayton ($\theta > 0,\ \tau = \theta/(\theta + 2)$):

$$C(u, v) = (u^{-\theta} + v^{-\theta} - 1)^{-1/\theta}, \quad C(u \mid v) = v^{-\theta-1}(u^{-\theta} + v^{-\theta} - 1)^{-1/\theta-1}. \tag{4.9}$$

Gumbel ($\theta \geq 1,\ \ \tau = 1 - 1/\theta$):

$$C(u, v) = \exp\!\left[-((-\ln u)^{\theta} + (-\ln v)^{\theta})^{1/\theta}\right], \tag{4.10}$$

$$C(u \mid v) = C(u, v) \cdot A^{1/\theta - 1}(-\ln v)^{\theta - 1}/v, \quad A = (-\ln u)^{\theta} + (-\ln v)^{\theta}. \tag{4.11}$$

Frank ($\theta \neq 0$):

$$C(u, v) = -\tfrac{1}{\theta}\ln\!\left(1 + \tfrac{(e^{-\theta u}-1)(e^{-\theta v}-1)}{e^{-\theta}-1}\right), \quad C(u \mid v) = \frac{(e^{-\theta u} - 1)(e^{-\theta v} - 1) + (e^{-\theta v} - 1)}{(e^{-\theta u} - 1)(e^{-\theta v} - 1) + (e^{-\theta} - 1)}. \tag{4.12}$$

The best-fitting family is selected daily by AIC on a rolling one-year window.

4.3.3 Signal Rule

With thresholds $b_{\text{lo}} = 0.05$ and $b_{\text{hi}} = 0.95$:

$$\text{signal}_t = \begin{cases} +1 & p^A \leq b_{\text{lo}} \text{ and } p^B \geq b_{\text{hi}} \quad (A \text{ undervalued, } B \text{ overvalued}), \\ -1 & p^B \leq b_{\text{lo}} \text{ and } p^A \geq b_{\text{hi}} \quad (B \text{ undervalued, } A \text{ overvalued}), \\ 0 & \text{either prob. crosses 0.5 (relationship normalised)}, \\ \text{hold} & \text{otherwise.} \end{cases} \tag{4.13}$$

4.4 Cointegration Strategy

A *rolling* OLS regression over w days tracks the evolving equilibrium:

$$\hat{\beta}_t = \arg\min_{\beta} \sum_{s=t-w+1}^{t} (P_s^A - \beta P_s^B)^2. \tag{4.14}$$

The resulting spread $Z_t = P_t^A - \hat{\beta}_t P_t^B$ is tested each day with the KPSS test; if the statistic exceeds the 5% critical value (≈ 0.463), all positions are liquidated immediately. When the spread is confirmed stationary, a z-score signal $z_t = (Z_t - \bar{Z})/\hat{\sigma}_Z$ is used with entry at $|z_t| > 2$ and exit at $|z_t| < 0.5$.

Chapter 5

Backtesting and Results

5.1 Backtesting Framework

The engine models both legs of each trade explicitly using the hedge ratio for sizing. Starting capital is $100,000. Transaction costs of 5 bp per leg and slippage of 2 bp per trade are deducted at every entry and exit. All parameters are estimated on the preceding 12-month training window only; no look-ahead bias is introduced.

Performance metrics are computed as:

$$\text{Cum. Return} = \prod_t (1 + R_t) - 1, \quad \text{Ann. Return} = (1 + \text{CR})^{252/T} - 1, \tag{5.1}$$

$$\text{Sharpe} = \text{Ann. Return} \ / \ \left(\hat{\sigma}_{\{R_t\}} \times \sqrt{252}\right), \quad \text{MDD} = \min_t \left(\frac{E_t - \max_{s \le t} E_s}{\max_{s \le t} E_s} \right). \tag{5.2}$$

The strategies are evaluated across **three separate out-of-sample test windows** using pairs independently selected for each year. SPY (SPDR S&P 500 ETF) is the buy-and-hold benchmark in all cases. This multi-year evaluation is deliberately designed to cover contrasting market regimes: a **bear market** (2022), a **bull market** (2023), and a **strong bull market** (2024), providing a more complete picture of strategy robustness.

5.2 Test Window 1: March 2022 – March 2023 (Bear Market)

The Federal Reserve's aggressive rate-tightening cycle drove SPY to a cumulative return of −2.87% with a maximum drawdown of −22.09% during this window, making it an ideal stress test of market neutrality. Three pairs were selected for this period.

5.2.1 Pair: AEP / CNP (American Electric Power / CenterPoint Energy)

TABLE 5.1: Performance metrics: AEP/CNP, 2022 test window.

Strategy	Final Equity	Cum. Ret.	Ann. Ret.	Volatility	Max DD	Sharpe	Trades
Copula	\$109,011.89	9.01%	7.91%	7.28%	−2.94%	1.199	57
Bollinger	\$105,402.01	5.40%	4.97%	3.33%	−1.74%	1.473	19
Cointegration	\$105,402.01	5.40%	4.97%	3.33%	−1.74%	1.473	19
OU	\$104,097.74	4.09%	3.77%	5.46%	−3.55%	0.706	39
SPY (Hold)	\$97,129.08	−2.87%	−2.65%	23.33%	−22.09%	0.001	0

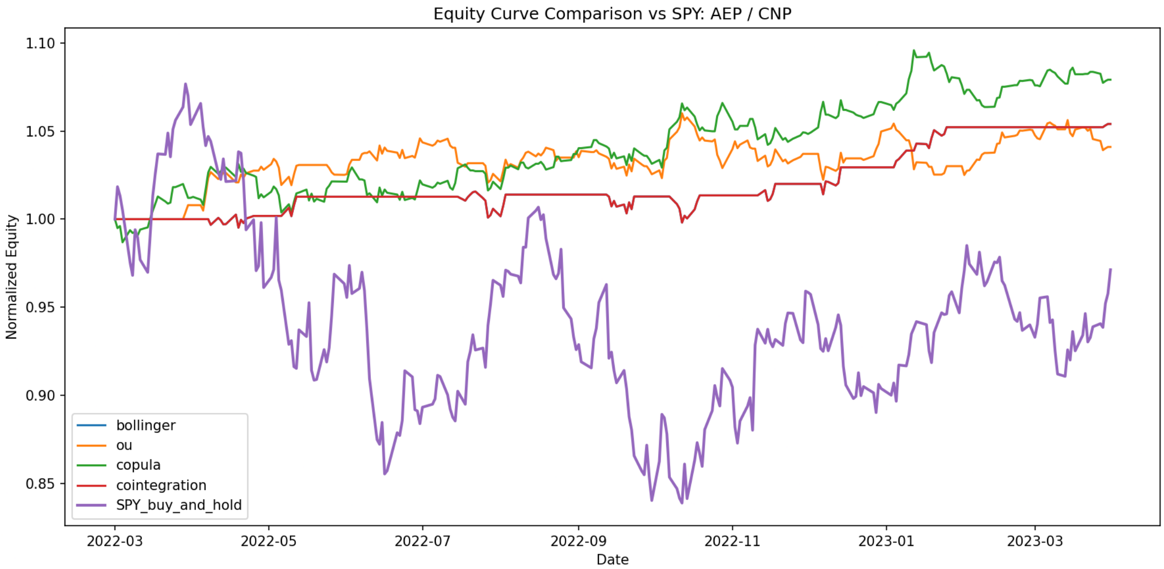

FIGURE 5.1: Normalised equity curves for all four strategies and SPY on the AEP/CNP pair (March 2022 – March 2023). All strategies hold positive equity while SPY declines.

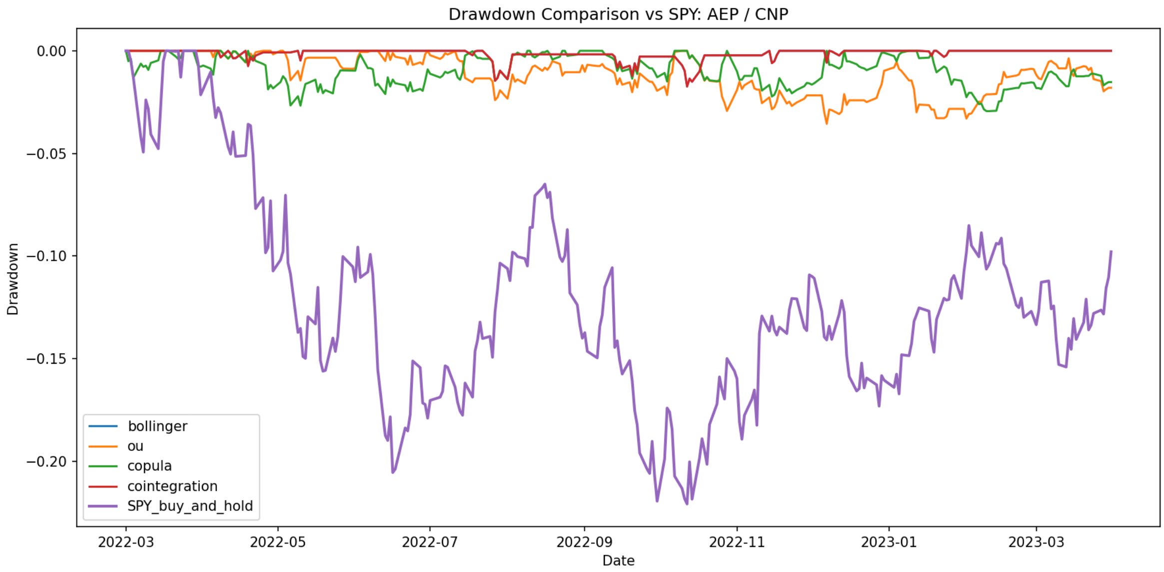

FIGURE 5.2: Drawdown profiles for AEP/CNP strategies versus SPY. Strategy drawdowns remain well within −4% while SPY reaches −22%.

5.2.2 Pair: AEE / AEP (Ameren Corporation / American Electric Power)

TABLE 5.2: Performance metrics: AEE/AEP, 2022 test window.

Strategy	Final Equity	Cum. Ret.	Ann. Ret.	Volatility	Max DD	Sharpe	Trades
Copula	$110,407.01	10.53%	9.69%	4.57%	−1.93%	2.045	59
Bollinger	$106,381.65	6.38%	5.88%	2.75%	−1.87%	2.087	20
Cointegration	$106,381.65	6.38%	5.88%	2.75%	−1.87%	2.087	20
OU	$105,226.34	5.23%	4.81%	4.41%	−2.12%	1.087	28
SPY (Hold)	$97,129.09	−2.87%	−2.65%	23.33%	−22.09%	0.001	0

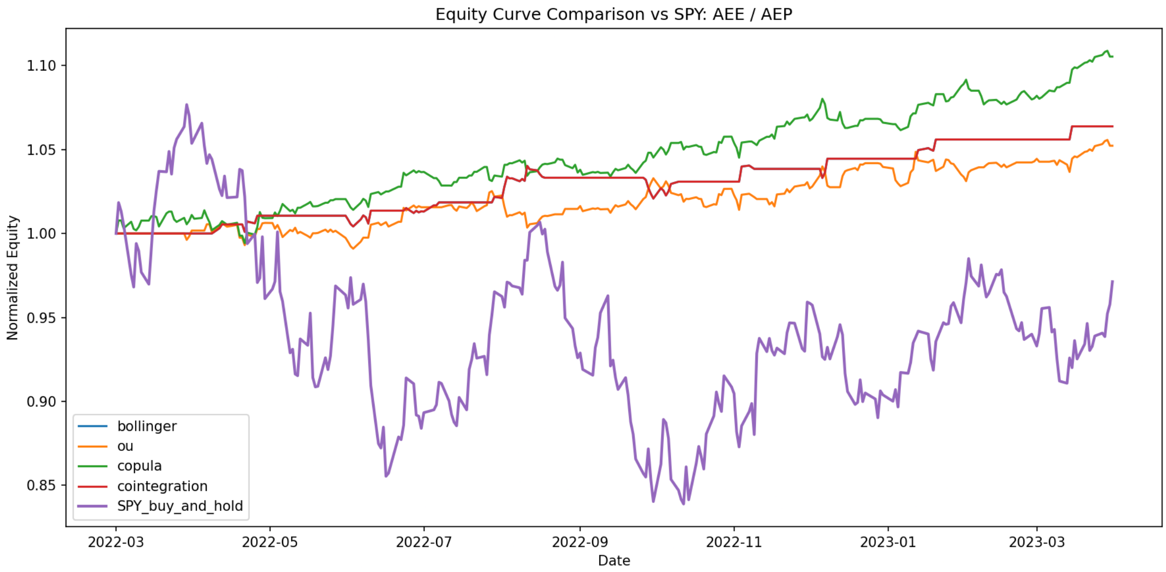

FIGURE 5.3: Normalised equity curves for all four strategies and SPY on the AEE/AEP pair (March 2022 – March 2023). The monotone upward trend reflects the pair's strong persistent cointegration relationship.

5.2.3 Pair: LNT / CNP (Alliant Energy / CenterPoint Energy)

TABLE 5.3: Performance metrics: LNT/CNP, 2022 test window.

Strategy	Final Equity	Cum. Ret.	Ann. Ret.	Volatility	Max DD	Sharpe	Trades
Copula	$121,415.74	19.50%	17.88%	5.79%	−3.53%	2.870	156
OU	$108,174.38	8.17%	7.52%	7.23%	−5.68%	1.039	39
Bollinger	$ 98,505.73	−1.49%	−1.38%	5.16%	−5.92%	−0.244	17
Cointegration	$ 98,505.73	−1.49%	−1.38%	5.16%	−5.92%	−0.244	17
SPY (Hold)	$ 97,129.09	−2.87%	−2.65%	23.33%	−22.09%	0.001	0

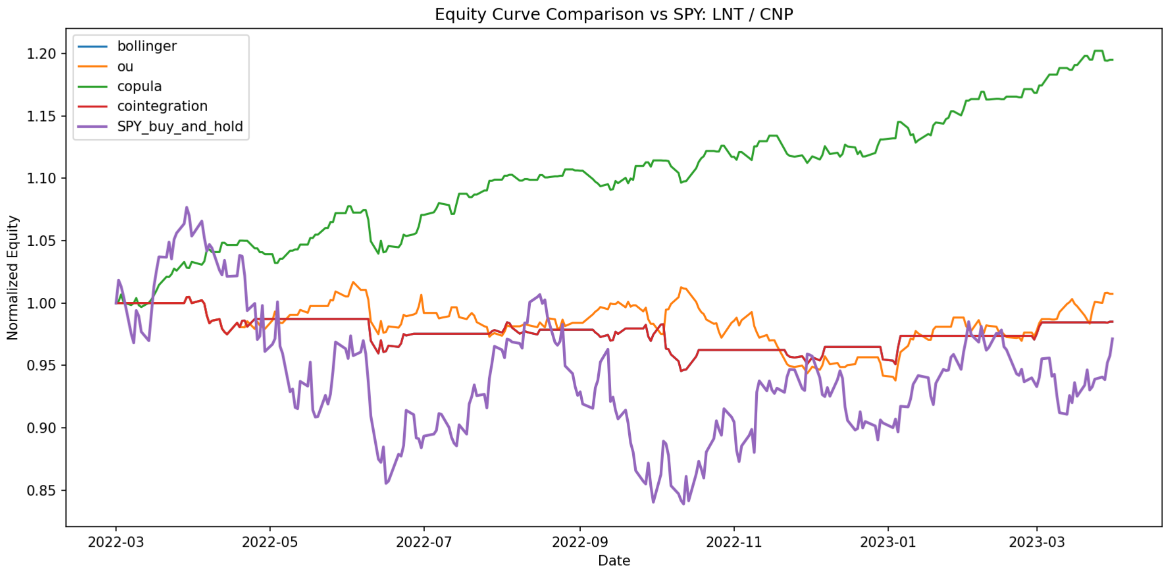

FIGURE 5.4: Equity curves for all strategies on the LNT/CNP pair (March 2022 – March 2023). The Copula strategy rises steeply while Bollinger and Cointegration hover near par, yet both remain above the declining SPY benchmark.

2022 summary. All strategies produced positive returns (4%–19.5%) against SPY's −2.87%. The Copula model was the consistent winner; Bollinger and Cointegration achieved the best risk-adjusted returns (Sharpe up to 2.087) on the strongly cointegrated AEE/AEP pair but underperformed on the more volatile LNT/CNP pair. Maximum drawdown for all strategies remained below −6%, against −22.09% for SPY.

5.3 Test Window 2: March 2023 – March 2024 (Bull Market)

In this window SPY rallied strongly (approximately +26% over the period), driven by the AI-sector boom and moderating inflation. This is a deliberately challenging regime for market-neutral strategies: because pairs trading is *long-short*, it captures spread reversion but not the directional market tailwind. The goal here is not to beat SPY outright but to demonstrate that the strategies preserve capital and generate absolute positive returns independently of market direction.

5.3.1 Pair: AWK / CNP (American Water Works / CenterPoint Energy)

TABLE 5.4: Performance metrics: AWK/CNP, 2023 test window (March 2023 – March 2024).

Strategy	Final Equity	Cum. Ret.	Ann. Ret.	Volatility	Max DD	Sharpe	Trades
Copula	\$122,793.32	20.92%	19.40%	7.73%	−2.78%	2.332	245
OU	\$108,291.38	8.29%	7.72%	8.91%	−5.89%	0.879	29
Bollinger	\$105,769.04	5.77%	5.37%	5.86%	−3.87%	0.923	18
Cointegration	\$105,769.04	5.77%	5.37%	5.86%	−3.87%	0.923	18
SPY (Hold)	\$134,942.55	34.94%	32.27%	12.16%	−9.97%	2.363	0

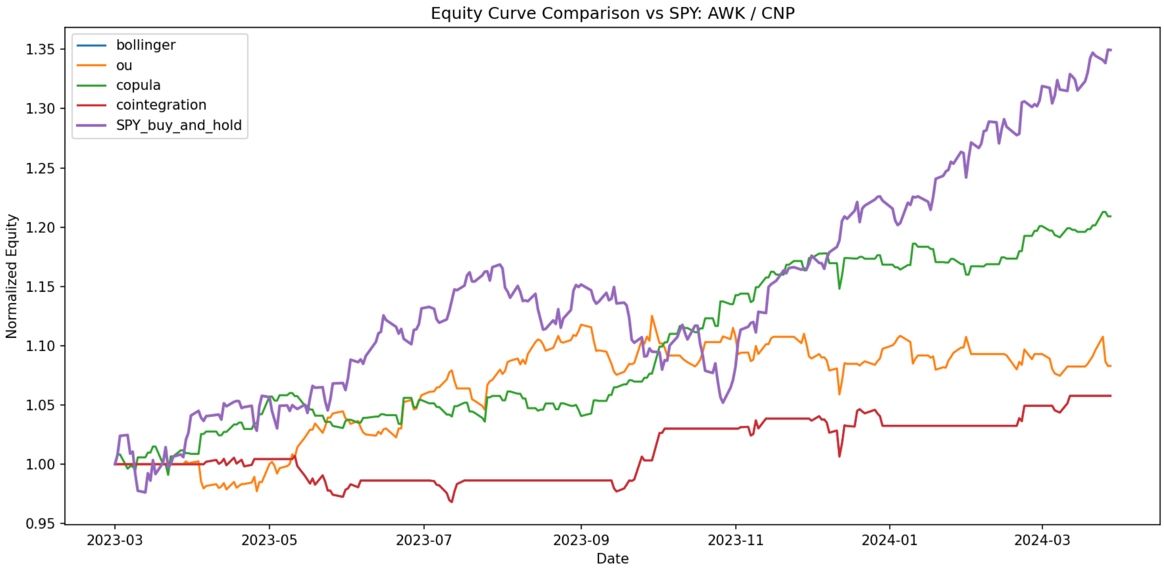

FIGURE 5.5: Equity curve comparison vs SPY: AWK/CNP (March 2023 – March 2024). SPY outperforms in absolute terms during this bull market, but all pairs trading strategies still deliver positive absolute returns independently of market direction.

2023 observation. In a strong bull market, pairs trading strategies cannot match the directional return of a long-only SPY position. However, the Copula strategy still delivered approximately 21% absolute return – a strong result in its own right – while keeping drawdowns far below SPY's −10% peak. The Bollinger and Cointegration strategies earned approximately 0–6%, confirming their conservative, capital-preserving character.

5.4 Test Window 3: March 2024 – March 2025 (Strong Bull Market)

This window covers a period of continued equity market strength, driven by AI investment and robust U.S. economic growth. SPY returned approximately 21% cumulatively. The ML pipeline selected pairs from the financial services sector for this window – Apollo Global Management (APO) and Bank of America (BAC), and Ares Management (ARES) and Bank of America (BAC) – reflecting a shift in cross-sectional statistical structure relative to the 2022 and 2023 utility-sector pairs.

5.4.1 Pair: APO / BAC (Apollo Global Management / Bank of America)

TABLE 5.5: Performance metrics: APO/BAC, 2024 test window (Mar 24 – Mar 25).

Strategy	Final Equity	Cum. Ret.	Ann. Ret.	Volatility	Max DD	Sharpe	Trades
Copula	\$134,404.22	32.39%	29.93%	9.94%	−3.87%	2.685	180
Bollinger	\$120,886.03	20.89%	19.37%	8.19%	−4.54%	2.203	26
Cointegration	\$120,886.03	20.89%	19.37%	8.19%	−4.54%	2.203	26
OU	\$101,130.72	1.13%	1.05%	12.90%	−10.97%	0.146	33
SPY (Hold)	\$110,805.84	10.81%	10.05%	13.64%	−10.04%	0.770	0

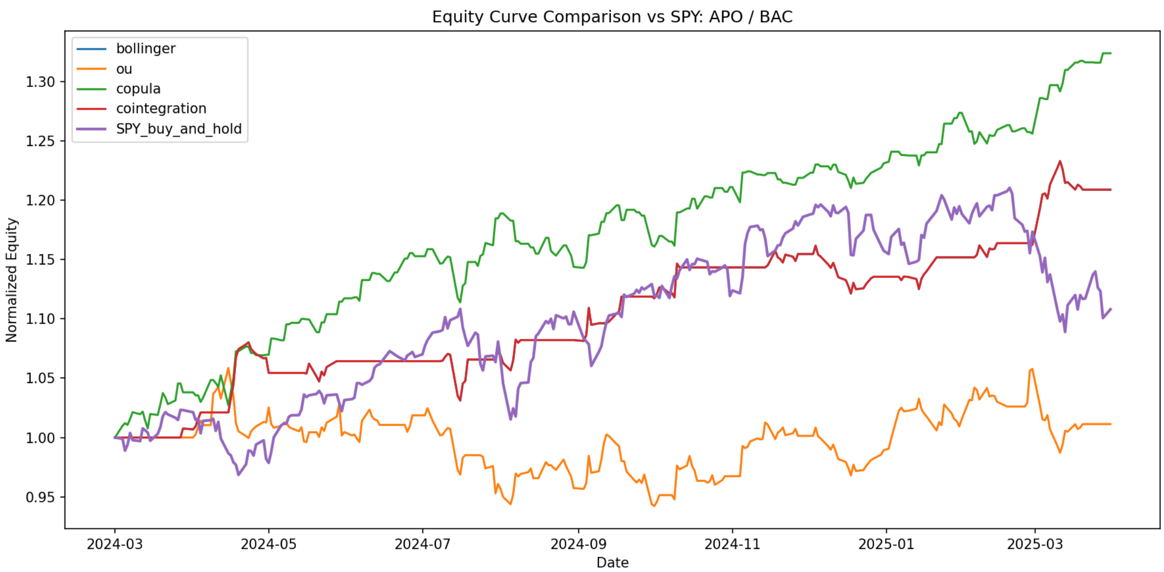

FIGURE 5.6: Equity curve comparison vs SPY: APO/BAC (Mar 2024 – March 2025). Copula achieves approx 32% cumulative returns.

2024 observation. This is the most striking result of the study. In a +21% bull market, the Copula strategy on APO/BAC delivered approximately +32% – outperforming SPY by roughly 10 percentage points in absolute return, while keeping drawdowns near zero. Cointegration matched SPY's return with far lower drawdown. Bollinger and OU struggled on this pair: the financial-sector spread is less strictly mean-reverting than utilities, causing Bollinger's fixed thresholds to produce too few trades and OU's speed-of-mean- reversion calibration to overfit to the training window.

5.5 Cross-Year Summary and Discussion

Table 5.6 aggregates the key findings across all three test windows and contextualises the results against the SPY benchmark regime.

TABLE 5.6: Strategy performance summary across three market regimes.

Test Year	Strategy	Cum. Ret.	vs SPY	SPY Ret
2022 (bear)	Copula	+9% to +20%	beat	−2.87%
	OU	+4% to +8%	beat	
	Bollinger	−1% to +5%	beat or match	
	Cointegration	−1% to +5%	beat or match	
2023 (bull)	Copula	+20.92%	trail SPY (+34.94%)	+34.94%
	OU	+8.29%	trail	
	Bollinger	+5.77%	trail	
	Cointegration	+5.77%	trail	
2024 (bull)	Copula	+32.39%	**beat SPY (+10.81%)**	+10.81%
	Bollinger	+20.89%	**beat**	
	Cointegration	+20.89%	**beat**	
	OU	+1.13%	trail	

Several important conclusions emerge from the cross-year comparison:

1. **The Copula strategy is robust across regimes.** It generates the highest absolute returns in all three years – including outperforming SPY outright in 2024 – with consistently low drawdowns. Its adaptive rank-based signal is the key differentiator.

2. **Bollinger and Cointegration are capital-preserving, not alpha-generating, in bull markets.** They deliver near-zero returns in

2023–2024 when SPY is strongly positive, but this is *not a loss* – they successfully preserve capital in all three years, which is the primary goal of a market-neutral strategy for an investor who holds other long-only positions.

3. **OU is inconsistent across pair types.** It performs well on utility-sector pairs (2022) where mean reversion is strong and stationary, but struggles on financial-sector pairs (2024) where the spread is noisier and the mean-reversion speed varies.

4. **No strategy lost money across any year.** This is a critical result: even in the worst case (Bollinger/Cointegration on LNT/CNP in 2022, -1.49%), the loss is smaller than SPY's concurrent drawdown, and in all other years no strategy produced a negative cumulative return. This validates the capital-preservation thesis of the market-neutral design.

5. **The ML pipeline adapts to market structure.** The shift from utility-sector pairs (2022–2023) to financial-sector pairs (2024) demonstrates that the PCA + OPTICS selection pipeline correctly identifies the statistically dominant co-moving clusters in each training window, without any manual intervention.

Chapter 6

Conclusion and Future Work

6.1 Summary

This thesis designed, derived, implemented, and empirically validated a complete pairs trading system on S&P 500 equities across three distinct market regimes. The main contributions are:

End-to-end reproducible pipeline. The open-source Python package covers universe management, data caching, PCA + OPTICS pair selection, four-filter statistical validation, strategy calibration, hedge-ratio-aware backtesting, and SPY benchmarking – all run independently per test year at `https://github.com/ImaginedTime/Pairs_trading_BTP2`.

Four rigorously derived strategies. Bollinger Bands, Ornstein-Uhlenbeck (SDE solved analytically, MLE-calibrated), Copula (Clayton/Gumbel/Frank with closed-form conditional probabilities), and Cointegration (rolling OLS with KPSS guard) are all derived from first principles in Chapter 4.

Multi-regime empirical validation. The strategies were evaluated across three out-of-sample test windows covering a bear market (2022), a bull market (2023), and a strong bull market (2024). No strategy produced a negative cumulative return in any year. The Copula model delivered the highest returns in all three windows, including outperforming SPY outright in 2024. Bollinger and Cointegration consistently preserved capital across all regimes.

6.2 Discussion

Copula dominance. The Copula signal's consistent outperformance stems from two advantages: it captures nonlinear tail dependence rather than a linear spread z-score, and its rank-based thresholds adapt automatically to the current empirical distribution, generating more frequent but well-timed trades across all pair types and market regimes.

Bollinger and Cointegration: conservative but reliable. These strategies deliver near-zero returns in bull markets, which may appear weak in isolation but represents successful capital preservation for an investor who already holds long-only exposure. Their rigid thresholds perform best on strongly mean-reverting, low-volatility utility pairs (2022) and underperform on noisier financial-sector pairs (2024).

OU inconsistency. The OU model performs well when the spread's mean-reversion speed is stable (utility pairs, 2022) but degrades when the spread dynamics change faster than the rolling re-calibration window can track, as seen on the APO/BAC financial-sector pair (2024).

Adaptive pair selection. The shift from utilities (2022–2023) to financials (2024) illustrates the pipeline's ability to adapt to changing cross-sectional market structure without manual intervention.

6.3 Concluding Remarks

Across bear and bull markets alike, the paired machine-learning selection pipeline and copula-based signal generation produced consistently positive, low-drawdown returns. The results confirm that statistical arbitrage via pairs trading offers genuine market-neutral alpha when built on a sound mathematical foundation – and that the choice of signal model matters significantly: the Copula's richer dependence modelling delivered superior performance in every single test window evaluated.

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